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Skin cancer detection using deep machine learning techniques

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ABSTRACT

Technological advancements have allowed people to have unfettered access to the internet from anywhere in the world. However, there is still little access to healthcare in rural and remote areas. This study highlights the potential of deep learning techniques in improving the early detection of skin cancer, a condition affecting millions globally. By addressing the challenges of class imbalance and dataset limitations, this research presents a model that can be integrated into digital health platforms, potentially saving lives by enabling earlier diagnosis and intervention, especially in underserved regions. The study also suggest using deep learning and few-shot learning when using machine learning techniques for skin cancer diagnosis. This study utilized a novel approach the use of raw images for training and test images for test data. These input images were then pre-processed using a deep model to identify and predict subsequent outputs using the model. In addition, the effect of the Convolutional Neural Network (CNN) effect in predicting accuracy using a skin lesion's texture to differentiate between benign and malignant lesions in the body was also examined using retrieved image elements from skin photos that were significant to skin cancer identification. The study focuses on using deep learning techniques to improve the detection of skin cancer from dermoscopic images. Deep learning a top-tier method for classifying skin lesions, was applied to create an end-to-end algorithm that could identify skin cancer more accurately. A variety of deep learning backbones were utilized, addressing the challenge of class imbalance in large datasets and seeking ways to boost performance even when only small datasets are available. To overcome these obstacles, the research leveraged transfer learning, data augmentation, and Generative Adversarial Networks (GANs). It further explored different sampling techniques and loss functions that could be effective for imbalanced datasets. The study also involved a comparison between ensemble models and hybrid models to determine which was more effective for the early detection of skin cancer. The paper concluded with a discussion of the challenges faced in the early detection of skin cancer, suggesting that while progress has been made, there are still significant hurdles to overcome.

1. Introduction

Cancer is caused by the growth of cells that have the ability to spread to other parts of the body [1]. Skin cancer is a highly malignant and perilous form of cancer. The only time skin cancer can be cured is when it is diagnosed in its early stages. The skin envelops all anatomical structures of the body, such as muscles and bones, and is indispensable to the human body. A minor modification in the functioning of the skin has a significant influence on every system in the body, establishing it as a prominent participant. A lesion area refers to the affected region of the skin. There is a wide range of different types of skin lesions. Each lesion is categorized into categories according to the specific type of skin cells from which it originated. Melanocytes generate melanocytic lesions, which bear resemblance to melanoma. Melanin is a protein pigment synthesized by melanocytes [1].

This study describes skin cancer and the challenges associated with its early detection. It explores the application of deep learning techniques, focusing on quantitative methods like few-shot learning and Convolutional Neural Networks (CNNs) to enhance accuracy. While a mixed-method strategy, combining both qualitative and quantitative approaches, could theoretically offer insights into the practical implementation and real-world challenges, the study focuses purely on quantitative methods due to the nature of machine learning models and the necessity for large-scale data-driven approaches.

Skin cancer, particularly melanoma, is one of the most aggressive forms of cancer and is a significant global health challenge. According to recent statistics from the World Health Organization (WHO), over 2–3 million non-melanoma skin cancers and 132,000 melanoma skin cancers

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occur globally each year [2]. Early detection is crucial in improving survival rates, but traditional diagnostic methods are often costly, time-consuming, and inaccessible, especially in remote areas [3]. The emergence of artificial intelligence (AI)-driven approaches, such as those described in this study, can provide an afford able, accurate, and accessible solution for early detection of skin cancer, which can potentially save lives. This research aims to develop deep learning models that can address the challenges of diagnosing skin cancer early, offering new opportunities for integration into real-world healthcare systems, such as telemedicine applications.

Non-melanocytic lesions arise from several types of skin cells, such as squamous or basal cells. The two categories of lesions can be visually differentiated by the presence or absence of several dermoscopic characteristics, such as pigment network. The subsequent stage involves identifying the nature of the lesion, namely whether it is a benign lesion or a malignant tumor. If the lesion falls into the latter category, it is then classified as one of the numerous skin lesions. When making these assessments, a set of dermoscopic features is also taken into account. Skin lesions are the main clinical markers of skin disorders such as basal cell carcinoma, melanoma, and seborrheic keratosis. As individuals grow older, they may develop benign skin growths known as basal cell carcinoma [4]. The two predominant types of skin cancer are non-melanoma and melanoma, both of which have a high fatality rate and are less prevalent. Melanoma is a malignancy that primarily impacts individuals of Caucasian descent. Typically, it impacts the torso of males and the lower extremities of females, however it can also manifest in other regions of the body [4].

More than twelve million individuals are affected by cancer, with a particular emphasis on the treatment of skin cancer in modern medicine [5]. Computer-aided detection [6] helps in the early identification of skin cancer. Medical professionals commonly employ the biopsy method to find and diagnose various illnesses. During the biopsy operation, the skin is removed or separated, and these skin samples undergo a series of laboratory testing. This operation is characterized by both discomfort and a significant investment of time. Depigmentation, characterized by the presence of scars, a blue-white veil, numerous blue-gray dots, pseudopods, multiple brown dots, globules, and pigmented networks, can develop in the skin [5]. Macroscopic images, also known as quantifiable images, are commonly captured using a standard digital camera or video. These images are frequently employed for computer analysis [6]. The clinical photographs exhibit several difficulties, such as poor image quality and the presence of artifacts such as skin lines, shadows, duplications, and hair. Due to these challenges, the study of cutaneous lesions is exceedingly complex [6].

Skin cancer, a condition defined by abnormal cell proliferation in human tissues, is widely recognized as a significant healthcare challenge for those affected by this ailment [7]. Considering the fact that the skin is the largest organ in the body, skin cancer is a prevalent and highly dangerous kind of cancer [8]. Skin cancer is the result of uncorrected DNA in skin cells [5,9]. There are two primary categories of skin cancer: melanoma and nonmelanoma skin cancers [6]. While the majority of skin cancer cases are nonmelanoma and have a low likelihood of spreading to other parts of the body, melanoma cancers are a rare, aggressive, and deadly kind of skin cancer that originates in melanocytes, which are cells found on the surface of the skin. Melanoma arises when melanocytes, the cells responsible for producing pigment, become cancerous. Pigment cells are most prominently apparent in the skin, however they can also be found in the eyes, other body regions, and various organs. The condition may manifest on the posterior, thorax, or visage. It occurs in the lower extremities of females. According to the American Cancer Society, melanoma skin cancer represents less than one percent of all reported occurrences. The condition is linked to an increased likelihood of death since it can only be effectively treated when identified at an early stage [10,11]. Therefore, it is crucial to discover the condition in the initial phase in order to increase the chances of a cure and minimize the expenses associated with therapy. As stated by Esteva [10].

The computational approach for identifying skin cancer involves several stages, including preprocessing, image segmentation, feature extraction, feature reduction, and classification [12,13]. This study presents a novel approach to few-shot meta-learning for skin cancer detection, utilizing a hybrid network composed of convolutional neural networks and prototype networks. The strategy is based on Kullback-Leibler divergence. The skin cancer classifications in the analysis of this study are displayed in Fig. 1.

Researchers have devised techniques for detecting skin cancer that can assist in promptly diagnosing the condition with precision and sensitivity, due to the urgent nature of the problem. Dermoscopy imaging is employed for the identification of melanoma in the human skin. Dermatologists analyze dermoscopic pictures to assess the presence of skin lesions. The field of dermatology, along with other medical disciplines, has greatly profited from advancements in the subfield of artificial intelligence known as computer vision. Computer vision aims to empower computer systems to extract information from digital images and automate tasks related to identifying objects, a capability that is inherent in human visual systems [6].

The performance of deep learning algorithms in computer vision tasks has improved since the introduction of the first convolutional neural network [14]. Recently, deep learning has been used in computer-aided skin cancer recognition tasks [14]. In the past ten years, the abundance of large amounts of labeled image data has significantly assisted deep learning algorithms in achieving comprehensive enhancements in computer vision tasks related to medical pictures [14]. Regrettably, deep neural networks trained on a little quantity of image data sometimes encounter failure as a result of overfitting. When there is just a limited amount of labeled picture data available for training, the ability to generalize successfully is reduced. Therefore, few-shot classification methods [15] have emerged to aid in learning from a limited quantity of labeled data. Few-shot learning methods enable computer vision models to quickly adapt to new tasks and situations using only a small number of training samples. Deep learning utilizes convolutional neural networks to acquire and identify highly distinctive characteristics from a dataset of training images, as a result of the challenges and inherent constraints of machine learning algorithms for the diagnosis of skin cancer. Every node in a Convolutional Neural Network (CNN) is linked to a restricted number of nodes in the subsequent layer. CNN has the ability to capture both the temporal and spatial properties of an image by utilizing the suitable filters [16]. This classifier utilizes a modified softmax function [16] to normalize input vectors into a probability distribution. The output values from the feature extractor are commonly inputted into a fully connected network. This study aims to develop an innovative machine-learning model for the diagnosis of skin cancer. It will utilize few-shot meta-learning and convolutional neural networks to quickly identify skin cancer.

The organization of this document is as follows. Section II focuses on the literature review and related work. The paradigm proposed for the detection of skin cancer is elucidated in Section 3. Section 4 presents the empirical results and performance evaluation of the classifiers utilized in this study. The conclusion is provided in Section 5.

1.1. Review of literature and relevant research

Utilizing deep learning techniques for the identification of skin cancer. While deep learning methods like CNNs have been successfully applied to skin cancer detection, they often struggle with the imbalance in skin lesion datasets, particularly in detecting rare but critical conditions like melanoma. Furthermore, their reliance on large datasets makes them less applicable in scenarios where data is scarce. This research addresses these shortcomings by utilizing GANs to generate synthetic data, improving model performance even in the face of small or imbalanced datasets.

Deep learning algorithms play a vital role in identifying and

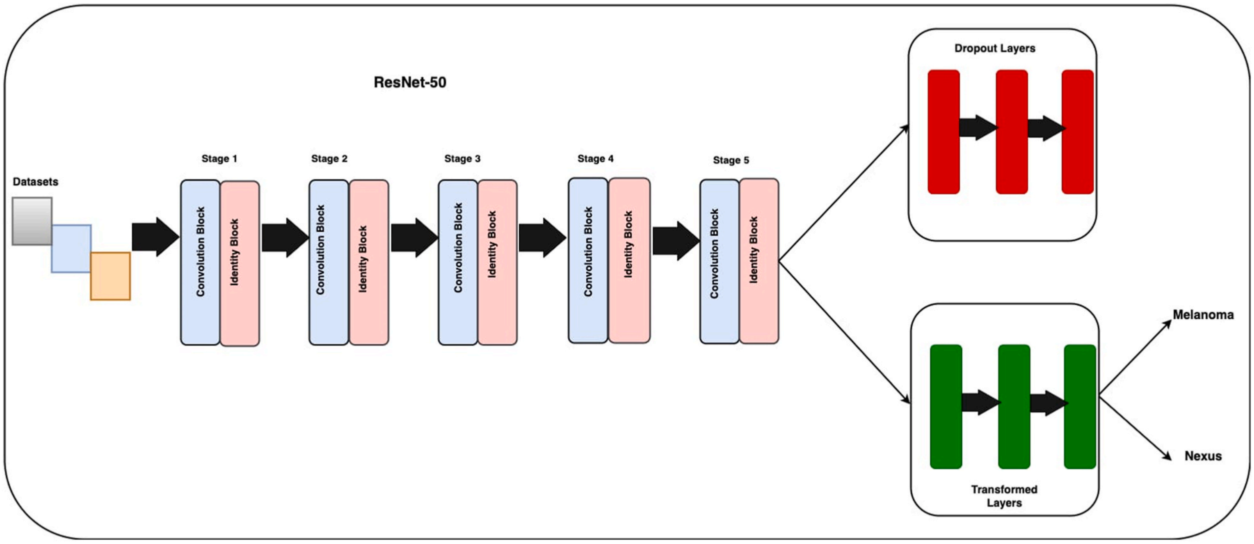

Fig. 1. Photographs depicting healthy skin and photographs depicting abnormal skin conditions.

diagnosing skin cancer. Their nodes are connected. The neural interconnection bears similarity to the structure of the brain. Together, their nodes work together to solve problems. Through the process of training neural networks for certain activities, they develop a high level of proficiency in that particular domains. We employed neural networks to train with the aim of categorizing photographs and distinguishing between different forms of skin cancer. The ISIC dataset showcases several types of skin lesions, as illustrated in Fig. 2. We conducted a study on various learning approaches, including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Kohonen Self-Organizing Neural Network (KNN), and Generative Adversarial Network (GAN), with the aim of detecting skin cancer. This article offers a thorough examination of each distinct deep neural network. Deep neural networks (DNNs) assist in the detection of skin cancer. Their nodes are connected. The neural interconnectedness exhibits a similarity to the architecture of the brain. Together, their nodes work together to solve problems. Through the process of training neural networks for certain activities, they develop a high level of proficiency in that particular fields. We employed neural networks to construct models for the classification of images and the discrimination of different types of skin cancer. The ISIC dataset showcases several types of skin lesions, as illustrated in Fig. 2. A study was done to investigate several learning algorithms, including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), K-Nearest Neighbors (KNN), and Generative Adversarial Networks (GAN), with the aim of detecting skin cancer. This article offers an extensive examination of each specific deep neural network.

Utilizing Artificial Neural Network (ANN) methods for the detection of skin cancer.

Artificial neural networks utilize statistical and nonlinear prediction techniques. It originates from the human brain. An Artificial Neural Network (ANN) comprises three layers of neurons. The neurons in the input layer propagate signals to the second or intermediate layer. The intermediate layers are hidden. Artificial neural networks (ANNs) often

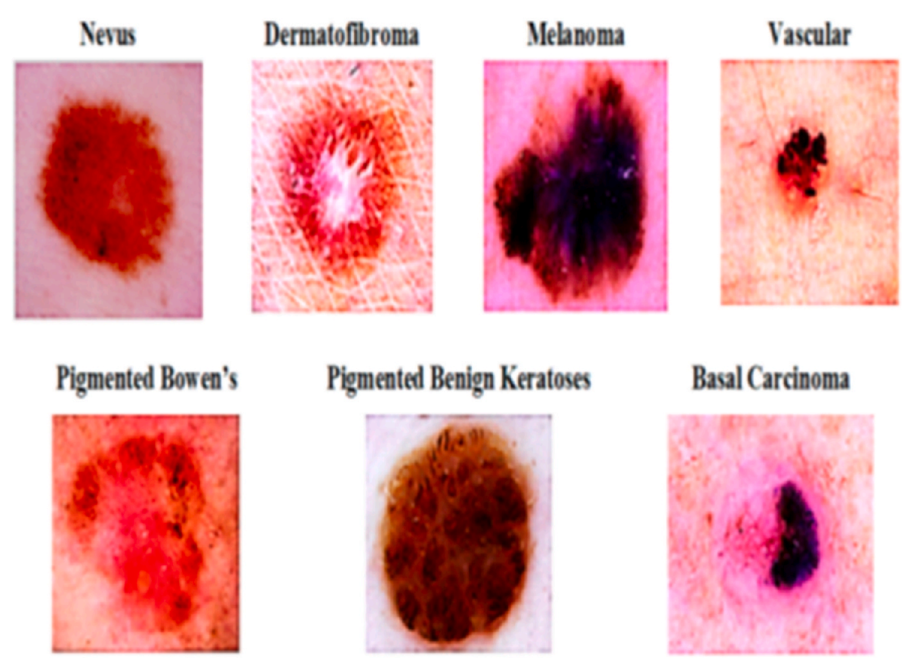

Fig. 2. displays the various classifications of skin diseases as presented in the source.

comprise multiple hidden layers. Intermediate neurons relay impulses to output neurons in the third layer. Backpropagation is utilized to obtain information about computations at each layer in order to manage complex input-output connections, similar to the functioning of a neural network. Neural networks and artificial neural networks are regarded as synonymous words in the realm of computer science. Fig. 3 illustrates the structure of the Artificial Neural Network (ANN). Fig. 2 illustrates the several categories of skin diseases in the ISIC dataset, as mentioned in the source [17,18]. Artificial neural networks employ non-linear and statistical methods to identify skin cancer. It originates from the human brain. An Artificial Neural Network (ANN) comprises three layers of neurons. The neurons in the input layer convey messages to the second or intermediate layer. The intermediate layers are hidden. Artificial neural networks (ANNs) often comprise multiple hidden layers. Intermediate neurons relay information to output neurons in the third layer. The backpropagation algorithm is utilized to learn the computations at each layer, enabling the neural network to effectively handle complex input-output relationships. Neural networks and artificial neural networks are synonymous words in the realm of computer science. Fig. 3 illustrates the structure of the Artificial Neural Network (ANN).

Artificial neural networks (ANNs) are used in skin cancer detection systems to classify extracted features. Upon the complete completion of training and classification of the training set, the input photographs are divided into two groups: melanoma or nonmelanoma. The quantity of input photographs has a direct influence on the quantity of hidden layers in an Artificial Neural Network (ANN). The input dataset establishes a linkage between the initial input layer and the hidden layer in the Artificial Neural Network (ANN) process. The decision to employ either a supervised or unsupervised learning approach for dataset analysis is contingent upon whether the dataset is labeled or unlabeled, correspondingly. A neural network obtains the weights of each connection or link in the network by either feed-forward or backpropagation architecture. Both algorithms employ the same underlying dataset, but they do so by identifying different patterns. Feed-forward neural networks propagate information only in a unidirectional fashion. Input layer exclusively receives data flows and transmits them to the output layer.

Xie et al. [19] Presented a technique for classifying skin lesions, categorizing them into two main groups: benign and malignant. The suggested system operates in three discrete steps. At the outset, the process of extracting lesions from images was carried out using a self-generating neural network (NN). During the second step, a basic artificial neural network (ANN) structure was utilized to extract features including the tumor border, texture, and color information, as depicted in Fig. 3 [17]. Artificial neural networks (ANNs) are used in skin cancer detection systems to classify extracted characteristics.

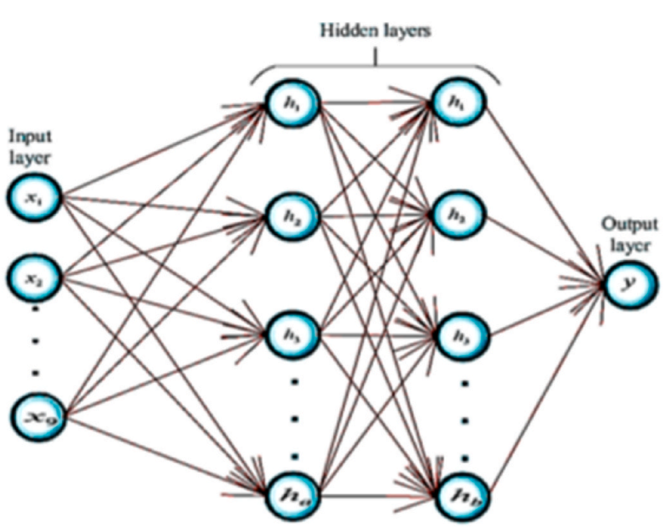

Fig. 3. depicts the typical arrangement of an artificial neural network.

After the training and classification of the training set is completed successfully, the input images are classified into two categories: melanoma or nonmelanoma. The quantity of input photographs has a direct influence on the quantity of hidden layers in an Artificial Neural Network (ANN). The input dataset establishes a link with the hidden layer in the initial stage of the Artificial Neural Network (ANN) process. The dataset can be analyzed using either a supervised or unsupervised learning approach, depending on its classification. The dataset may potentially be devoid of labels. A neural network employs either a feed-forward architecture or backpropagation algorithm to obtain the weights associated with each network connection or link. Every architecture in the underlying dataset utilizes a unique pattern. Neural networks built with a feed-forward design transmit information only in one way.

The papers listed employ Artificial Neural Networks (ANN) with the Backpropagation algorithm as the training algorithm for identifying skin cancer. The initial investigation utilized a dataset including 31 dermascopic pictures, with feature extraction performed using the ABCD parameters. The study reached a level of accuracy of 96.9 %. For the second investigation, a total of 90 dermoscopic images were utilized, and the feature extraction process involved the utilization of the maximum entropy method. The study did not provide explicit details about the precise outcomes attained.

1.2. Thresholding and gray-level manipulation

Co-occurrence matrix.

The ANN with Backpropagation algorithm achieved an accuracy of 86.6 % for feature extraction [20]. The classification task involved distinguishing between malignant and noncancerous conditions. The dataset consisted of 31 dermascopic pictures, and the 2D-wavelet transform was employed for feature extraction.

Feature extraction and thresholding, when used to the term "segmentation," pertain to the procedure of dividing or separating something into distinct segments or sections. In this particular context, the term "NIL" serves as an abbreviated form of "no information available." The numeral " [21]" could potentially serve as a citation or origin. Lastly, the term "malignant" is actually synonymous with "benign". This study employed a feed-forward artificial neural network (ANN) with the backpropagation training strategy to categorize 326 lesion photos. The classification relied on the discriminant qualities of the tumor, including its color and form characteristics. The achieved accuracy was 80 % [22]. The categorization of malignancy/non-malignancy was conducted using a backpropagation neural network as the classifier. For this aim, a grand total of 448 photos with mixed types were utilized. The segmentation process utilized the techniques of Region of Interest (ROI) and Spatial Relationship Matrix (SRM).

The ANN model's accuracy in differentiating between malignant and noncancerous patients is 70.4 % [23].

The backpropagation algorithm was employed to evaluate a dataset consisting of 30 photographs, which were categorized as either malignant or noncancerous. The photos were analyzed using RGB color features and GLCM algorithms to extract pertinent properties. The precision attained with this methodology was 86.6 %, as documented in reference [24]. This approach is frequently employed in comparable investigations.

The study employed a feed-forward backpropagation neural network to examine 200 dermoscopic images for the identification of moles and melanoma. The features were obtained using the ABCD rule.

The artificial neural network achieves a classification accuracy of 97.51 % in discriminating between malignant and noncancerous cells [25].

1.3. Using the backpropagation algorithm

The study utilized an algorithm to examine and analyze 50 dermascopic photographs. The GLCM method was employed to extract characteristics from the photos. The algorithm's accuracy was determined to be 88 %. The study incorporated the utilization of an artificial neural network (ANN) to categorize 180 skin lesion photos into BCC/non-BCC classifications. The technique of histogram equalization was employed to improve the contrast of the photos. The Artificial Neural Network (ANN) achieved a classification accuracy of 93.33 % in this task. A separate investigation was conducted to categorize skin lesions as either melanoma or nonmelanoma using an Artificial Neural Network (ANN).

The Levenberg–Marquardt method (LM), the durable Backpropagation (RBP), and the developed scaled conjugate algorithm were utilized for the analysis of 135 lesion pictures using gradient-based learning (GCG) algorithms. Several classifiers were employed to avoid misclassification. The classifiers were evaluated for accuracy, with SCG achieving a 91.9 % accuracy rate and LM achieving …

The RBP value for the meta-ensemble is 95.1, whereas the ANN value is 88.1. The categorization is limited to either malignancy or benignancy.

A model consisting of a Backpropagation Neural Network (BPN)

The study employed a fuzzy neural network to examine datasets containing information on the Caucasian race and the xanthous-race. The neural network utilized an autonomous mechanism to extract lesions. The system attained a level of accuracy of 94.17 %.

The sensitivity is 95 % and the specificity is 93.75 %. Information is transmitted solely from the input layer to the output layer.

Xie et al. [19] Proposed a methodology for classifying skin lesions into two primary categories: benign and malignant. The functioning of the proposed system necessitated three distinct steps. Initially, the lesions were extracted from the photos using a self-generating neural network. During the second phase, characteristics such as the tumor borders, texture, and color details were isolated and analyzed. The system extracted a total of 57 features, including seven novel features that pertain to descriptions of lesion borders. To choose the best set of features, one can use principal component analysis (PCA) to reduce the number of dimensions of the features. To classify lesions, a neural network ensemble model was used in the final stage. The ensemble neural network integrates backpropagation (BP) neural networks and fuzzy neural networks in order to enhance classification accuracy. The classification results of the proposed system were compared to those of different classifiers, including SVM, KNN, random forest, Adaboot, and others. The proposed model exhibited higher performance in comparison to the prior classifiers, with a sensitivity improvement of at least 7.5 percent and an accuracy rate of 91.11 percent.

The skin cancer diagnostic method developed by Masood et al. [19] Employs artificial neural networks (ANN) for the purpose of automation. This study also investigated the effectiveness of three artificial neural network (ANN) learning techniques, specifically scaled conjugate gradient (SCG) [24], and robust backpropagation (RP). [26], and Levenberg-Marquardt (LM) [27]. After conducting a performance comparison, it was found that the LM algorithm achieved the highest specificity score of 95.1 %. Furthermore, the system consistently demonstrated its ability to effectively diagnose benign lesions. In contrast, the SCG learning algorithm demonstrated excellent performance when the number of epochs was increased, leading to a sensitivity value of 92.6 percent. The proposed mole classification system aims to improve the timely detection of melanoma skin cancer [20]. The feature extraction of the suggested system followed the ABCD rule of lesions. The term "ABCD" denotes the attributes of asymmetry, border irregularity, color variation, and diameter size when characterizing a mole. The Mumford-Shah and Harris Stephen algorithms were employed to assess the asymmetry and delineations of a mole. In the suggested methodology, melanoma is defined as any mole colors other than black, cinnamon, or brown, as these colors are commonly observed in normal moles. The threshold for detecting melanoma was established at a diameter above 6 mm, as melanoma moles frequently exhibit this size. By employing a backpropagation feed-forward artificial neural network (ANN), the suggested system attained a classification accuracy of 97.51 % in categorizing moles into three distinct groups: common, uncommon, and melanoma.

Fig. 4 depicts a potential automated approach for diagnosing skin cancer [28], which utilizes backpropagation neural networks. This system employed the 2D-wavelet transform technique to extract distinctive characteristics. The proposed artificial neural network (ANN) model categorizes input photos into two distinct classes: malignant and noncancerous. Choudhari and Biday [23] suggested creating an extra artificial neural network (ANN) based system for diagnosing skin cancer. The photos were segmented using a thresholding approach based on maximum entropy. The utilization of a gray-level co-occurrence matrix (GLCM) was employed to extract distinguishing features of skin lesions. The feed-forward artificial neural network (ANN) accurately classified the input photos into either a benign or malignant stage of skin cancer, achieving an accuracy score of 86.66 percent.

Aswin et al.'s [24] The method for identifying skin cancer is based on the utilization of artificial neural networks (ANN) and genetic algorithms (GA), which are unique techniques. The Dull-Rozar application, a specialized tool for medical imaging, was employed to preprocess the pictures with the aim of hair removal. The Otsu thresholding method was employed to isolate the area of interest (ROI). In addition, the GLCM method was used to extract several characteristics from the segmented images. The classification of lesion images into malignant and noncancerous categories was accomplished by employing a combined Artificial Neural Network (ANN) and Genetic Algorithm (GA) classifier. The proposed method attained a comprehensive accuracy rate of 88 percent. The information in Table 1 presents a thorough overview of different artificial neural network (ANN) methods used for identifying skin cancer.

1.4. Methods for identifying skin cancer via Convolutional Neural Networks (CNNs)

A convolutional neural network (CNN) is a fundamental and extensively utilized type of deep neural network in the field of computer vision. It is used to detect and categorize input images and photographs. CNN is an invaluable resource for obtaining and examining data from both local and worldwide sources. It accomplishes this by employing less intricate curves and edges to create intricate elements like forms and portions [32]. The hidden layers of CNN include of convolutional layers, non-linear pooling layers, and fully connected layers [33]. CNNs typically consist of multiple convolutional layers followed by several fully linked layers. The creation of a Convolutional Neural Network (CNN) involves three primary layers: convolution, pooling, and full connection [34]. What is the essence of something? Fig. 5 depicts the architecture of a Convolutional Neural Network (CNN).

Automated deep learning systems employing convolutional neural networks (CNN) have demonstrated exceptional performance in the domains of medical imaging detection, segmentation, and classification [36,18]. Lequan et al. [37] Proposed a state-of-the-art Convolutional Neural Network (CNN) with advanced capabilities to accurately identify melanoma. In the segmentation phase, performance was improved by utilizing a fully convolutional residual network (FCRN) consisting of 16 residual blocks. The proposed approach entails the classification of data by computing the mean of the support vector machine (SVM) and softmax classifiers. The classification of melanoma attained an accuracy of 85.5 % when segmentation was used, and 82.8 % when segmentation was not used. DeVries and Ramachandram [38] Suggested a multi-scale convolutional neural network (CNN) utilizing an inception v3 deep neural network that has been trained on the ImageNet dataset. The skin cancer classification process entailed enhancing the pre-existing Inception v3 model by incorporating two distinct resolution scales for the

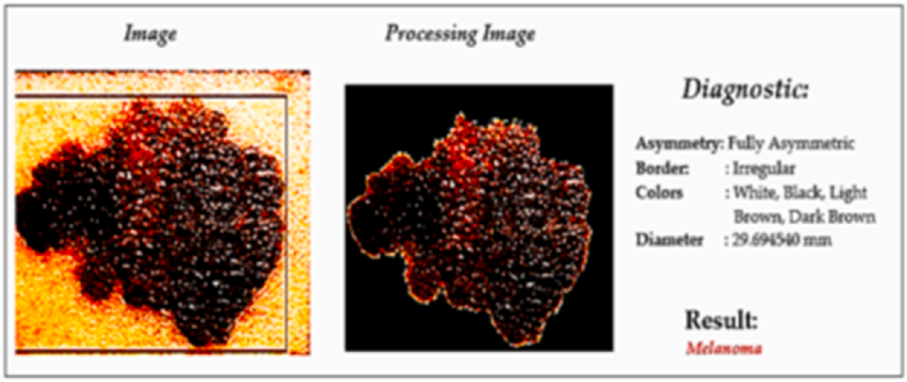

Fig. 4. illustrates the application of Artificial Neural Networks (ANN) in the detection of Skin Cancer.

Table 1
Comparative analysis of Artificial Neural Network (ANN) for skin cancer detection.

Reference	Skin Cancer Diagnoses	Training Algorithm	Dataset Used	Description	Resuls (%)
[29]	Melanoma	ANN with Backpropagation algorithm	31 dermascopic images	ABCD Parameters for Feature extraction	Accuracy (96.9)
[30]	Melanoma/ Nonmelanoma	ANN with Backpropagation algorithm	90 dermascopic images	maximum entropy for thresholding, and graylevel co-occurrence matrix for features extraction	Accuracy (86.6)
[31]	Cancerous/ noncancerous	ANN with Backpropagation algorithm	31 dermascopic images	2D-wavelet transform for feature extraction and thresholding for segmentation	NIL

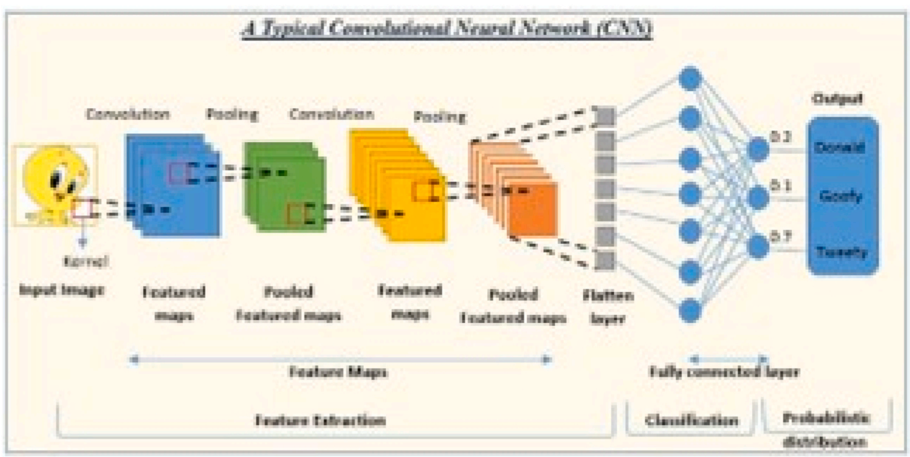

Fig. 5. illustrates the fundamental structure of a Convolutional Neural Network (CNN) as described in Ref. [35].

input lesion images: a larger-scale and a smaller-scale. The morphology of the lesions and their overall contextual information were documented using the coarse scale. On the other hand, a greater amount of specific textual data was collected regarding the lesion in order to distinguish it from other forms of skin lesions.

The method proposed by Mahbod et al. [39] The process of identifying skin disorders involves extracting significant features from well-established, pre-trained deep Convolutional Neural Networks (CNNs). The study included deep feature generators, namely pre-trained models including AlexNet, ResNet-18, and VGG16. The purpose of deploying these models was to extract features, which were subsequently utilized to train a multi-class SVM classifier. The classifier scores were ultimately combined to achieve classification. The suggested method demonstrated a high level of accuracy in identifying melanoma and seborrheic keratosis (SK) using the ISIC 2017 dataset, achieving an AUC performance of 97.55 % and 83.83 % respectively. A proposed model for categorizing 12 different types of skin lesions utilized a convolutional neural network (CNN) with the pre-trained ResNet-152 [40]. The model was first trained with 3797 images of lesions, along with an extra 29 augmentations generated through adjustments in scale and lighting positions. The suggested technique attained an area under the curve (AUC) of 0.99 for the classification of hemangioma lesions, pyogenic granuloma (PG) lesions, and intraepithelial carcinoma (IC) skin lesions.

Dorj et al. [41] Presented a methodology for classifying photos of four distinct types of skin lesions. The challenge utilized a classifier that employed an error-correcting output coding Support Vector Machine (SVM) after conducting feature extraction using a pre-trained deep Convolutional Neural Network (CNN) called AlexNet. The proposed approach received the greatest mean scores for sensitivity, specificity, and accuracy in detecting SCC, actinic keratosis (AK), and BCC. The readings were 95.1 %, 98.9 %, and 94.17 %, respectively. Kalouche [42] Proposed a pre-trained deep convolutional neural network (CNN) architecture called VGG-16, comprising of five convolutional blocks and three final fine-tuned layers. Fig. 6 depicts the VCG-16 model that has been suggested. The VCG-16 models demonstrated a classification accuracy of 78 % in appropriately identifying lesion images as melanoma skin cancer. A convolutional neural network (CNN) was suggested as an approach to identify the boundaries of skin lesions in photographs. The deep-learning method was developed using a dataset including 1200 photos of healthy skin and 400 shots of skin lesions. The proposed methodology yielded an accuracy rate of 86.67 %, effectively categorizing the input photographs into two primary groups: normal skin images and lesion images. Table 2 displays a comprehensive compilation of skin cancer screening methods that utilize CNN classifiers.

1.5. Kohonen Self-Organizing Neural Network (KNN)-based skin cancer detection techniques

The Kohonen self-organizing map is a highly esteemed variant of deep neural networks. Convolutional Neural Networks (CNNs) are trained using unsupervised learning, which allows a K-Nearest Neighbors (KNN) algorithm to learn without human interaction and with limited understanding of the input data's properties. A KNN generally comprises of two layers. The two levels on the two-dimensional plane are known as the competitive and input layers, respectively. The connections between these two fully connected layers are made unidirectionally from the first layer to the second layer dimension. Applying the KNN algorithm for clustering data is possible, even when the connections between the input data elements are unknown. It may alternatively be called a self-organizing map. Since each node in the competitive layer also serves as the output node, KNNs do not have a separate output layer.

In essence, a KNN algorithm reduces the dimensionality of the data. High-dimensional data can be transformed into a lower dimension, such as a two-dimensional plane. As a result, it offers several forms of input dataset representation. K-nearest neighbors (KNNs) differ from other neural networks in their use of competitive learning rather than feed-forward learning or error correction-based learning, as seen in back-propagation networks (BPN). The K-nearest neighbors (KNN) approach maintains the topological configuration of the input data space while decreasing the dimensionality from high to low. The term "preservation" denotes the act of sustaining a consistent spatial separation between data points. This strategy utilizes a method in which data points that are close to each other in the input data space are given a shorter mapping distance, while data points that are far apart are given a longer mapping distance, based on their relative distance. Hence, the optimal tool for examining data with a large number of dimensions is the KNN (K-Nearest Neighbors) algorithm. The generalization capacity of a KNN is a crucial characteristic that allows the network to accurately identify and categorize new input that it has not encountered before. Fig. 7 depicts the architecture of a KNN (K-Nearest Neighbors) algorithm. One major advantage of a KKN is its capacity to precisely depict intricate relationships among data points, including non-linear correlations. The advantages of KNNs have led to the incorporation of these algorithms in skin cancer detection systems.

Lenhardt et al. [36] Proposed a skin cancer detection methodology that use the K-Nearest Neighbors (KNN) algorithm. The suggested methodology entails using synchronous fluorescence spectra acquired from nevus, melanoma, and normal skin samples to educate neural networks. Fluorescence spectra were acquired from human patients directly after surgical removal by collecting samples and analyzing them using a fluorescence spectrophotometer. The observed spectra were subjected to dimensionality reduction using the PCA technique. After completing the training procedure, the melanoma detection capabilities of KNN (K-Nearest Neighbors) and ANN (Artificial Neural Network) were compared. The artificial neural network (ANN) exhibited a classification error ranging from 3 to 4 percent in the test dataset, whereas the k-nearest neighbors (KNN) demonstrated a classification error of 2–3 percent. An approach has been suggested to use a mix of self-organizing neural networks and radial basis function neural networks (RBF) for the diagnosis of three distinct kinds of skin cancer: basal cell carcinoma (BCC), melanoma, and squamous cell carcinoma (SCC) [37,18]. The prescribed methodology was employed to analyze lesion images and extract color, GLCM, and morphological data. These retrieved features were then entered into the categorization model. Furthermore, the efficacy of the suggested system's classification was evaluated in comparison to that of ANN, naïve-Bayes, and k-nearest neighbor classifiers. The proposed methodology attained an accuracy score of 93.150685, surpassing the performance of the k-nearest neighbor (71.232877), artificial neural networks (63.013699), and naïve Bayes (56.164384) models.

Sajid et al. [38] introduced a different automated approach using K-nearest neighbors (KNN) to detect skin cancer. The proposed technique utilized a median filter to effectively diminish noise. The pictures were segmented using an algorithm that employed statistical region growth and merging after filtering. This system employed a wide variety of textual and statistical elements. Textual features were derived from a curvelet domain, whereas statistical features were obtained from lesion images. The proposed method achieved a classification accuracy of 98.3 % in accurately differentiating between malignant and noncancerous input pictures. This study included a range of classifiers, including Support Vector Machines (SVM), Backpropagation Neural Networks (BPN), and 3-layer Neural Networks (NN). Subsequently, the performance of each classifier was evaluated and compared to that of the suggested approach. The proposed approach achieved a maximum accuracy of 98.3 % in detecting skin cancer, outperforming SVM (91.1 %), BPN (90.4 %), and 3-layer NN (90.5 %) in terms of accuracy. Table 3 displays data on skin cancer diagnosis methods that employ the KNN algorithm.

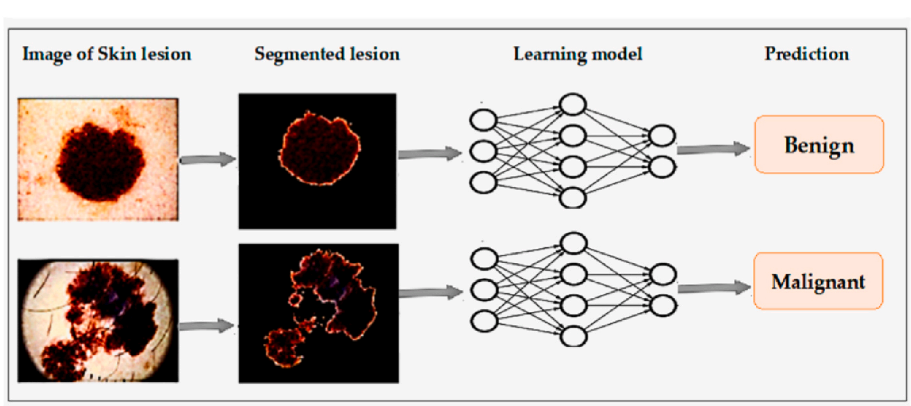

Fig. 6. Implementation of skin cancer diagnosis by CNN [1].

Table 2
A comparative analysis of CNN Techniques to detect skin cancer.

Ref.	Skin Cancer Diagnoses	Training Algorithm	Dataset Used	Description	Results (%)
[4]	Benign/malignant	LightNet (deep learning framework), used for classification	ISIC 2016 dataset	Fewer parameters and well suited for mobile applications	Accuracy (81.6), sensitivity (14.9), specificity (98)
[5]	Melanoma/benign	CNN classifier	170 skin lesion images	Two convolving layers in CNN	Accuracy (81), sensitivity (81), specificity (80)
[6]	BCC/SCC/melanoma/AK	SVM with deep CNN	3753 dermoscopic images	Pertained to deep CNN and AlexNet for features extraction	Accuracy (SCC: 95.1, AK: 98.9, BCC: 94.17)
[7]	Melanoma/benign Keratinocyte carcinomas/ benign SK	Deep CNN	ISIC-Dermoscopic Archive	Expert-level performance against 21 certified dermatologists	Accuracy (72.1)
[8]	Malignant melanoma and BC carcinoma	CNN with Res-Net 152 architecture	The first dataset has 170 images the second dataset contains 1300 images	Augmentor Python library for augmentation.	AUC (melanoma: 96, BCC: 91)
[10]	Melanoma/nonmelanoma	SVM-trained, with CNN, extracted features	DermIS dataset and DermQuest data	A median filter for noise removal and CNN for feature extraction	Accuracy (93.75)
[12]	Malignant melanoma/ nevus/SK	CNN as a single neural net architecture	ISIC 2017 dataset	CNN ensemble of AlexNet, VGGNet, and GoogleNetfor classification	Average AUC:9 84.8), average accuracy (83.8)
[14]	BCC/nonBCC	CNN	40 FF-OCT images	Trained CNN, consisting of 10 layers for feature extraction	Accuracy (95.93), sensitivity (95.2), specificity (96.54)
[15]	Cancerous/noncancerous	CNN	1730 skin lesions and background images	Focused on edge detection	Accuracy (86.67)
[1]	Benign/melanoma	VGG-16 and CNN	ISIC dataset	The dataset was trained on three separate learning models	Accuracy (78)
[16]	Benign/malignant	CNN	ISIC dataset	ABCD symptomatic checklist for feature extraction	Accuracy (89.5)
[17]	Benign/malignant	Region-based CNN with ResNet152	2742 dermoscopic images from ISIC dataset	The region of interest was extracted by mask and region-based CNN, then ResNet152 is used for classification.	Accuracy (90.4), sensitivity (82), specificity (92.5)
[19]	Melanoma/non melanoma	ResNet-50 with deep transfer learning	3600 lesion images from the ISIC dataset	The proposed model showed better performance than o InceptionV3, Densenet169, Inception ResNetV2, and Mobilenet	Accuracy (93.5), precision (94) recall (77), F1_ score (85)
[43]	BCC/non-BCC	Pruned ResNet18	297 FF-OCT images	K-fold cross-validation was applied to measure the performance of the proposed system	Accuracy (80)
[28]	AK/melanocytic nevus/ BCC/SK/SCC	CNN	1300 skin lesion images	Mean subtraction for each image pooled multi-scale feature extraction process and pooling in augmented-feature space	Accuracy (81.8)
[20]	Benign/malignant	A very deep residual CNN and FCRN	ISIC 2016 database	FCRN incorporated with a multi-scale contextual information integration the technique was proposed for accurate lesions segmentation	Accuracy (94.9), sensitivity (91.1), specificity (95.7), Jaccard index (82.9), dice coefficient (89.7)
[28]	Benign/malignant	CNN with 5-fold cross-validation	1760 dermoscopic images	Images were preprocessed on the basis of melanoma cytological findings	Accuracy (84.7), sensitivity (80.9), specificity (88.1)
[21]	Melanoma/SK	Deep multi-scale CNN	ISIC dataset	The proposed model used Inception-v3 model, which was trained on ImageNet.	Accuracy (90.3), AUC (94.3)
[22]	Compound nevus/ malignant melanoma	CNN	AtlasDerm, Derma, Dermnet, Danderm, DermIS and DermQuest datasets	BVLC-AlexNet model, trained from ImageNet the dataset was used for fine-tuning	Mean average precision (70)
[23]	Benign/malignant	Deep CNN	ISIC dataset	Data augmentation was performed for data balancing	Accuracy (80.3), precision (81), AUC (69)
[24]	melanoma/BCC/ melanocytic nevus/AK/benign keratosis/vascular lesion/dermatofibroma	CNN model with LeNet approach	ISIC dataset	The adaptive piecewise linear activation function was used to increase system performance	Accuracy (95.86)
[25]	Melanoma/BCC/ melanocytic nevus/Bowen's	CNN model with LeNet approach	ISIC dataset	The adaptive piecewise linear activation function was used	Accuracy (95.86)

(continued on next page)

Table 2 (*continued*)

Ref.	Skin Cancer Diagnoses	Training Algorithm	Dataset Used	Description	Results (%)
	disease/AK/benign keratosis/vascular lesion/dermatofibroma			to increase system performance	
[44]	Malignant melanoma/SK	SVM classification with features extracted with pre-trained deep models named AlexNet, ResNet-18, and VGG16	ISIC dataset	SVM scores were mapped to probabilities with logistic regression function for evaluation	Average AUC (90.69)
[26]	Melanoma/benign keratosis/melanocytic nevi/BCC/AK/IC/atypical nevi/dermatofibroma/ vascular lesions	Deep CNN architecture (DenseNet 201, Inception v3, ResNet 152 and InceptionResNet v2)	HAM10000 and PH2 dataset	Deep learning models outperformed highly trained dermatologists in overall mean results by at least 11 %	ROC AUC (DenseNet 201: 98.79–98.16, Inception v3: 98.60–97.80, ResNet 152: 98.61–98.04, InceptionResNet v2: 98.20–96.10)
[27]	Lipoma/fibroma/sclerosis/ melanoma	Deep region-based CNN and fuzzy C means clustering	ISIC dataset	Combination of the region-based CNN and fuzzy C-means ensured more accuracy in disease detection	Accuracy (94.8) sensitivity (97.81) specificity (94.17)
[32]	Malignant/benign	6-layers deep CNN	MED-NODE and ISIC datasets	Illumination factor in images affected the performance of the system	Accuracy (77.50)
[33]	Melanoma/non melanoma	The hybrid of fully CNN with autoencoder and decoder, and RNN	ISIC dataset	Proposed model outperformed state-of-art SegNet, FCN, and ExB architecture	Accuracy (98) Jaccard index (93), sensitivity (95), specificity (94)
[34]	Benign/malignant	2-layer CNN with a novel regularizer	ISIC dataset	Proposed regularization technique controlled complexity by adding a penalty on the dispersion value of the classifier's weight matrix	Accuracy (97.49) AUC (98), sensitivity (94.3), specificity (93.6)

CNN = Convolutional neural network; SVM = Support vector machine; ISIC = International skin imaging collaboration; HAM10000 = Human-against-machine dataset with 10,000 images; AK = Actinic keratosis; IC = Intraepithelial carcinoma; FF-OCT = Full-field optical coherence tomography; BCC = Basal cell carcinoma; SCC = Squamous cell carcinoma; FCN = Fully convolutional network; BVLC = Berkeley Vision and Learning Center; SK= Seborrheic keratosis; FCRN = Fully convolutional residual network.

1.6. *Generative Adversarial Network (GAN)-based skin cancer detection techniques*

Generative adversarial neural networks (DNNs) are a potent variant of DNNs that are influenced by the principles of zero-sum game theory [39]. The core notion of Generative Adversarial Networks (GANs) is rooted in the idea that two neural networks, specifically a discriminator and a generator, participate in a competitive process to identify and reproduce variations present in a given dataset. The generator module employs the data distribution to produce synthetic data samples with the purpose of misleading the discriminator module. On the other hand, the discriminator module's objective is to differentiate between authentic and misleading data samples [40]. Both of these neural networks do these operations repeatedly during the training phase, and with each iteration, their performance enhances. The main benefit of a GAN network is its capacity to produce synthetic samples that closely resemble genuine samples, while following the same data distribution, such as extremely lifelike photos. Moreover, it can effectively tackle a significant issue in deep learning, namely the limited availability of training instances. Scientists have employed various types of Generative Adversarial Networks (GANs), including Vanilla GAN, conditional GAN (CGAN), deep convolutional GAN (DCGAN), super-resolution GAN (SRGAN), and Laplacian Pyramid GAN (LPGAN). Currently, the utilization of Generative Adversarial Networks (GANs) has demonstrated significant efficacy in the detection and diagnosis of skin cancer. Fig. 8 depicts the structure of a Generative Adversarial Network (GAN).Both of these neural networks do these operations repeatedly during the training phase, and with each iteration, their performance enhances. The main benefit of a GAN network is its capacity to produce synthetic samples that closely resemble genuine samples, while following the same data distribution, such as extremely lifelike photos. Moreover, it can effectively tackle a significant issue in deep learning, namely the limited availability of training instances. Scientists have employed various types of Generative Adversarial Networks (GANs), including Vanilla GAN, conditional GAN (CGAN), deep convolutional GAN (DCGAN), super-resolution GAN (SRGAN), and Laplacian Pyramid GAN (LPGAN). Presently, the utilization of GANs has demonstrated to be quite beneficial in the detection of skin cancer. Fig. 8 depicts the structure of a Generative Adversarial Network (GAN).

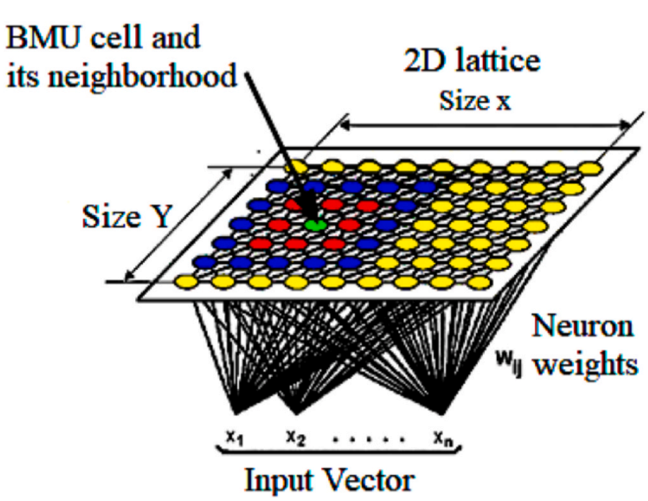

Fig. 7. Basic KNN structure [35], BMU= Best matching unit.

Rashid et al. [40] Developed a skin lesion categorization method

Table 3
A comparative analysis of KNN techniques to diagnosis skin cancer.

Reference	Skin Cancer Diagnoses	Training Algorithm	Dataset Used	Description	Results (%)
[37]	Cancerous/noncancerous	Modified KNN	500 lesion images	Automated Otsu method of thresholding for segmentation	Accuracy (98.3)
[36]	Melanoma/nevus/normal skin	SOM and feed-forward NN	50 skin lesion images	PCA for decreasing spectra's dimensionality	Accuracy (96–98)
[38]	BCC, SCC, and melanoma	SOM and RBF	DermQuest and Dermnet datasets	GCM morphological and colour features were extracted	Accuracy (93.15)

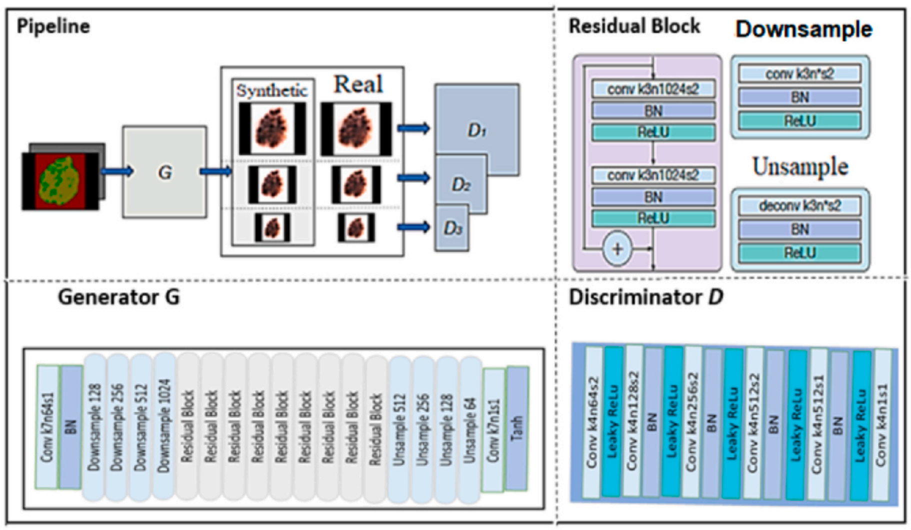

Fig. 8. Basic GAN architecture [39].

utilizing Generative Adversarial Networks (GAN). The proposed technique employed a Generative Adversarial Network (GAN) to produce synthetic skin lesion images that closely mimic authentic ones. These synthetic images were then used to supplement a training dataset of photographs. The discriminator module employed a Convolutional Neural Network (CNN) as a classifier, while the generator module utilized a deconvolutional network. CNN received instructions on seven distinct categorizations of skin lesions. The results of the suggested system were compared to those of ResNet-50 and DenseNet. DenseNet attained an accuracy rate of 81.5 %, ResNet-50 achieved an accuracy rate of 79.2 %, and the recommended technique achieved the best accuracy rate of 86.1 % for the classification of skin lesions. Deep learning methods achieve acceptable levels of accuracy, but they necessitate large, imbalanced, and uniform training datasets. Bisla et al. [12] proposed the utilization of Generative Adversarial Networks (GAN) To overcome the constraints of data purification, data augmentation is employed in deep learning. The proposed approach employed decoupled deep convolutional GANs to generate data. A more advanced and refined dataset was used to further strengthen a pre-trained ResNet-50 model. The improved model was subsequently employed to categorize dermoscopic pictures into three groups: nevus, seborrheic keratosis (SK), and melanoma. The proposed technique attained an accuracy of 86.1 %, exceeding the performance of the original ResNet-50 model.The Progressive GAN (PGAN) is an innovative technique for enhancing data by focusing on skin lesions, using self-attention. Furthermore, the application of the stabilizing technique led to an improvement in the generative model. The proposed system attained a precision of 70.1 %, surpassing the 67.3 % precision of the non-augmented system. Table 4 presents a collection of skin cancer detection methods that make use of Generative Adversarial Networks (GANs). The table provides details regarding the specific type of skin cancer that was detected, the classifier that was utilized, the dataset that was applied, and the matching outcomes that were achieved.

2. Methodology design and implementation

Machine learning (ML), a leading component of the ongoing artificial intelligence (AI) revolution, offers new opportunities for clinical

Table 4
Collection of skin cancer detection methods that use GAN techniques.

Reference	Skin Cancer Diagnoses	Training Algorithm	Dataset Used	Description	Results (%)
[40]	Melanoma/nevus/SK	Deep convolutional GAN	ISIC 2017, ISIC 2018, PH2	Decoupled deep convolutional GANs for data augmentation	ROC AUC (91.5), accuracy (86.1)
[41]	BCC/vascular/pigmented benign keratosis/pigmented Bowen's/nevus/dermatofibroma	Self-attention-based PGAN	ISIC 2018	A generative model was enhanced with a stabilization technique	Accuracy (70.1)
[42]	AK/BCC/benign keratosis/dermatofibroma/melanoma/ melanocytic nevus/vascular lesion	GAN	ISIC 2018	The proposed system used deconvolutional network and CNN as generator and discriminator module	Accuracy (86.1)

practice through the utilization of medical images [45]. Deep learning, is a specific type of machine learning that utilizes multiple layers of processing to effectively capture complex patterns and abstract concepts inside data [46]. Recently, deep learning algorithms have demonstrated exceptional performance in challenges related to the classification of skin cancer [47,2]. Convolutional neural networks (CNNs) are widely accepted as the prevailing method for addressing computer vision challenges in the business. Deep learning (DL) possesses the capacity to autonomously acquire potent feature representations and is more adaptable in accommodating diverse domains and applications in comparison to conventional machine learning methods. This obviates the necessity for intricate feature engineering [3]. The objective of this study was to create an automatic classification technique using dermoscopic images of various common skin cancers, employing a deep learning algorithm.

The article illustrates the detailed procedure of picture classification using deep learning, as shown in Fig. 9. Image preprocessing refers to the procedure of improving the quality of image data by removing undesirable distortions and enhancing significant visual components. This enhancement enables computer vision models to operate more effectively by leveraging enhanced data. The goal of model training is to detect the most prominent patterns existing in an image. These characteristics may be exclusive to a specific category and, hence, assist the model in distinguishing between various categories.

We conducted picture classification, which entails assigning a singular label or category to a set of photos. Each image is expected to be classified into a single category. The categorization model employs an algorithm that takes a photo as input and produces a prediction about the image's classification.

The skin illness photos were classified using deep learning by performing the subsequent steps.

- ➢ Conducting a thorough analysis, examination, and comprehension of the data
- ➢ Constructing an input pipeline
- ➢ Constructing and developing the model
- ➢ Model training
- ➢ Conducting tests and assessments on the model
- ➢ Enhancing the model and iterating the procedure

Selecting an appropriate backbone model is essential for obtaining acceptable experimental outcomes. Fig. 10 below illustrates a prominent and exceptional center structure that has remained significant over the years. The ILSVRC dataset, derived from the ImageNet dataset, has been extensively utilized from 2012 to 2021 for the classification and localization of images. The dataset comprises 100,000 test shots, 14,197,122 training photos, and 50,000 validation images. It encompasses a total of 1000 distinct varieties of things [45].

ImageNet is an extensive collection of actual photographs that have been annotated with comprehensive metadata. This dataset is widely utilized in contemporary computer vision research. In order to acquire a new skill, one can utilize a pre-trained image classification network that

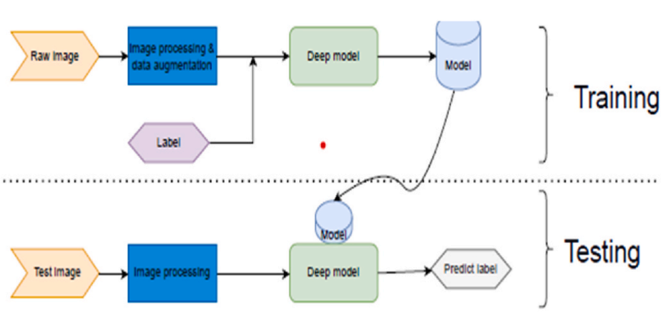

Fig. 9. Flow diagram of image classification based on deep learning

has previously acquired the capacity to extract valuable and potent features from real-life photos. Most pre-trained networks on ImageNet receive training. The curve in Fig. 10 displays the furthest level of accuracy attained in categorizing images on ImageNet from 2011 to 2022, surpassing the prior supremacy of CNNs prior to 2021.

Furthermore, the foundational architectures of neural networks are consistently updated. CoCa [46] achieved state-of-the-art (SOTA) top-one accuracy of 91.0 % on ImageNet in 2022, using a highly optimized encoder. Attention pooling was used to merge image-side data. Our study is guided by these algorithms and approaches, as mentioned earlier. Choosing a suitable backbone model is crucial for achieving favorable experimental results. Fig. 10 exemplifies the enduring prominence of the backbone throughout the years. The ImageNet Large Scale Visual Recognition Challenge (ILSVRC) dataset, covering the period from 2012 to 2021, is the most commonly used subset of ImageNet for the tasks of classifying and localizing images. The dataset comprises 100,000 test shots, 14,197,122 training photos, and 50,000 validation images. It encompasses a grand total of 1000 distinct categories of items [47]. ImageNet is a large compilation of real-world photos that have been annotated with additional information. This dataset is commonly used in modern computer vision research.

To provide a basis for acquiring a new skill, one can make use of a pre-trained image classification network that has already developed the capability to extract valuable and powerful features from real-life photographs. The majority of pre-trained networks on ImageNet are trained. The graph depicted in Fig. 10 displays the highest level of precision achieved in classifying images on ImageNet from 2011 to 2022, surpassing the previous dominance of CNNs before 2021. Furthermore, the fundamental structures of neural networks undergo constant updates. In 2022, CoCa [46] achieved state-of-the-art (SOTA) performance on ImageNet by utilizing a highly optimized encoder, resulting in a top-one accuracy of 91.0 percent.

Furthermore, the activation function was utilized to perform nonlinear operations on the input data of neurons, with the purpose of extracting significant feature information from the original input data. An activation function is a non-linear function. By augmenting the number of layers in the neural network model, we can get the most efficient feature information by means of many iterations and data training. Therefore, to accomplish feature classification, we utilized the Softmax function as the activation function in the neural network model and applied it in the output layer of the model [2]. The softmax function is commonly employed as a classifier. The calculating formula is as follows:

$$sk = \frac{exp(R_k)}{\sum_{k=1}^{n} exp(R_k)}$$

The above formula uses the variable n to represent the number of neurons in the current layer, and Rk represents the nonlinear transformation value for the k-th neuron in this layer. Softmax is a classification technique that can provide separate output categories based on the features. In Softmax, the value of each neuron can be understood as the probability that is linked to the relevant category. The formula for the Softmax function is illustrated in Fig. 11:

In addition, attention pooling was employed to merge the data from the image side. Our study is guided by these algorithms and approaches, as mentioned earlier.

2.1. Datasets

The progress of applying deep learning research in medicine has been limited due to the lack of dependable clinical datasets. Paper V provides a concise overview of twelve commonly used dermoscopy datasets that are employed for the classification of skin cancer. The International Skin Imaging Collaboration (ISIC) dataset has gained

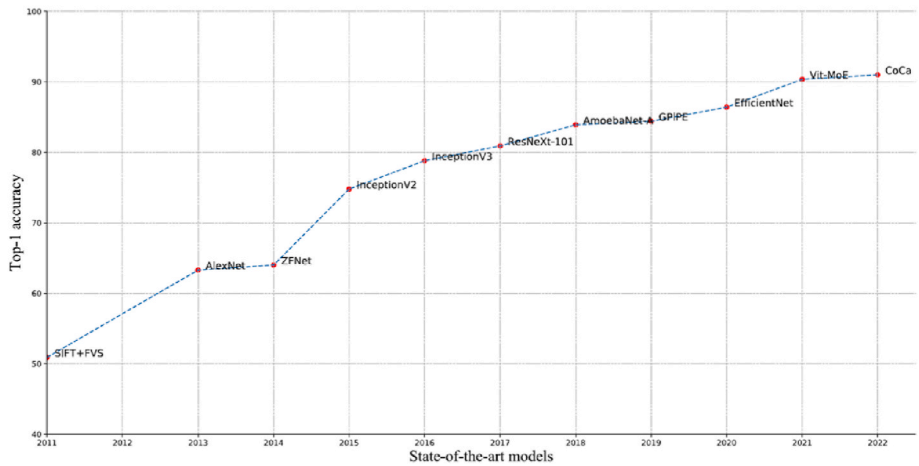

Fig. 10. Image classification: top one accuracy on ImageNet from 2011 to 2022 [45].

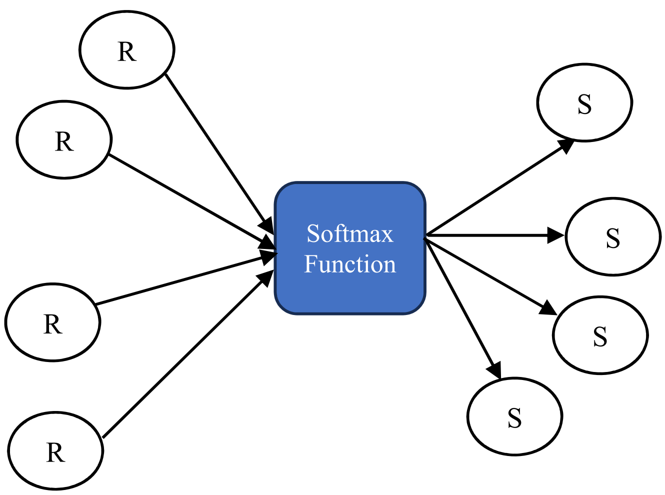

Fig. 11. Schematic diagram for calculating the Softmax function.

significant recognition since 2016. It has been employed in a yearly competition for categorizing skin cancer. In Paper VI, the ISIC-2018 dataset [46] was employed as the experimental object, while the lesion images were sourced from the HAM10000 dataset [47]. This study utilized the HAM10000 dataset as the training data for the experiment. Both of these databases consist of seven distinct categories of skin diseases and are completely overlapping.

For this study, we carefully chose 1000 photos from the ISIC-2017 dataset, with an equal number of benign and malignant images, to ensure that the categories were balanced. The dataset has been meticulously annotated by experts, with each item containing annotations of extraordinary quality. We utilized the Dermoscopy Atlas [2] dataset, which consists of one thousand publicly available dermoscopy images, to train and assess a Convolutional Neural Network (CNN) model for a seven-point checklist to determine the malignancy of skin lesions (see Fig. 12).

The 7-point checklist comprises the identification of seven dermoscopic criteria, which have been selected based on their strong link with melanoma. The seven aforementioned features comprise the chromatic qualities as well as the contour and/or texture of the lesion. A score is calculated by analyzing dermoscopy photographs of a melanocytic skin lesion to determine the presence of specific criteria. If the total score is three or more, The lesion is diagnosed as melanoma. Alternatively, it is regarded as harmless. According to the data presented in Table 5 and Fig. 13, three criteria that are considered "major" have been given a score of two each, while four criteria that are considered "minor" have been given a score of one each.

2.2. Preprocessing of the images

"Once garbage enters, it remains outside." This slogan is used frequently in machine learning fields, computer science, mathematics, and computer science. This statement emphasizes an important factor: if the input data is of low quality, even the most skilled computer vision models will show below-average performance. Consistently, it is vital to get high-quality facts for the present undertaking. Preprocessing is

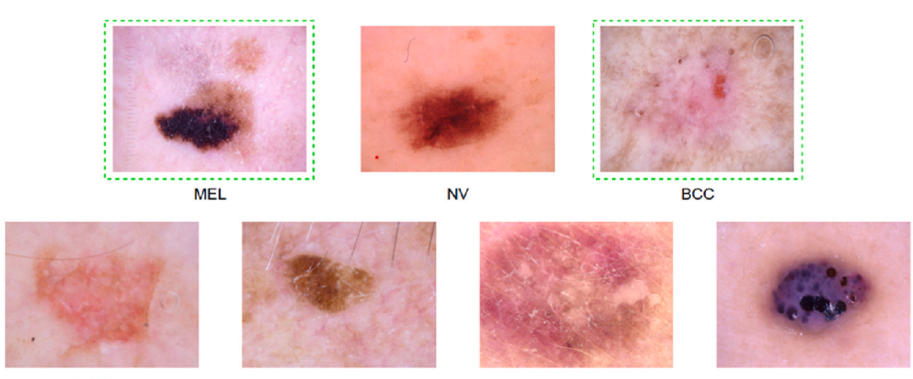

Fig. 12. Samples of each type of skin lesion from the ISIC 2018 dataset where BCC and MEL are skin cancers (in the green and dotted box) while others are common diseases [46] (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 5
The 7-point checklist criteria with related individual scores.

Clinical Features	Score
Major Criteria	
Pigment Network	2
Blue-Whitish Veil	2
Vascular Pattern	2
Minor Criteria	
Irregular Streak	1
Irregular Pigmentation	1
Irregular dots	1
Structures of Regression	1

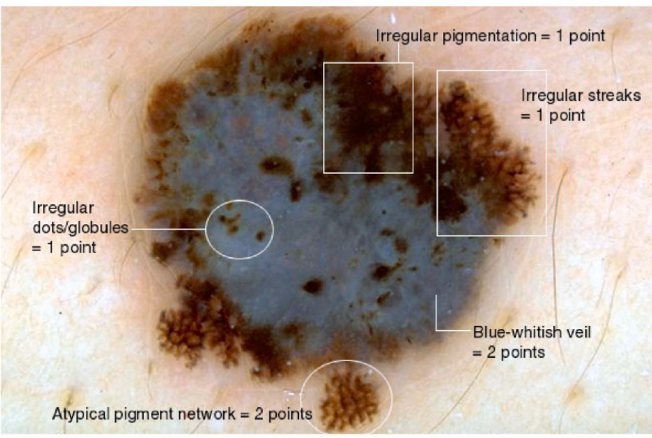

Fig. 13. A diagnostic example of a 7-point checklist, different structures are present- and a score of seven is reached [47].

essential for attaining optimal results, even when using high-quality data.

Obtaining an image is the fundamental basis for image classification. There are many possible reasons of dataset bias in skin lesion imaging. Some imaging procedures or technologies may generate biases in measurements. One type of bias that can greatly hinder the effectiveness of automated diagnosis for clinical purposes is the incorporation of medical interventions in the data. An instance of this phenomenon is the presence of misleading connections in images of skin abnormalities due to mistakes in labeling.

Images commonly undergo a procedure known as picture preprocessing before being utilized for model training and inference. This may involve altering alterations to the hue, proportions, or alignment, among other potential options. Preprocessing is performed to improve the quality of the image, hence facilitating a more effective analysis. Preprocessing facilitates the removal of undesirable distortions and enhances key qualities that are essential for the present application I am working on. Efficient picture preprocessing requires the use of software to achieve best performance and desired outcomes.

There are several justifications for performing image preprocessing. Firstly, the model's fully connected layers necessitate that all images are represented as arrays of identical dimensions. Preprocessing the model can accelerate model inference and reduce the duration of model training. Rescaling sizable input images will greatly expedite the training process without substantially impacting the model's performance.

2.3. The augmentation of data

The evaluation of overfitting and underfitting is conducted during the experiment by analyzing the training and test errors. The training error steadily decreases over the repeated process of training the deep neural network model. However, it rises again when the test error reaches a specific level. The character "*b*" represents an inflection point in Fig. 14. The neural network model is considered to be underfitted if it has not reached the optimal state prior to the occurrence of this inflection point. After reaching the *b* inflection point, the test error starts to rise while the training error continues to decrease. This disparity arises due to the model's insufficient capacity to generalize and the subpar performance of the test set, despite the training set fitting remarkably well. Consequently, the model might be categorized as overfitting. Utilizing data augmentation can significantly enhance the performance of a network model in cases where it struggles to accurately fit the data. This is accomplished by incorporating fabricated data into the model (see Fig. 15).

Data augmentation is a technique that involves modifying photos to generate multiple variations of the same content. This process is beneficial as it expands the pool of training samples accessible for the model. When applying arbitrary modifications to the brightness, rotation, or scale of an input image, a model must take into account the appearance of the image subject under different circumstances.

Data augmentation techniques involve many methods of image preparation. Nevertheless, there exists a significant distinction between picture preprocessing techniques and image augmentation. Both the training and test sets undergo picture preprocessing, whereas image augmentation is exclusively done to the training dataset. Therefore, in certain situations, a conversion that has the capacity to function as an improvement technique may be more effectively utilized as an initial measure.

Capturing an image that encompasses all conceivable real-world scenarios is unattainable. In this context, augmentation can provide assistance. By augmenting the images, we may expand the size of the training data set and incorporate additional instances that may be challenging to encounter in real-life scenarios. By extending the current training data to various contexts, the model gains knowledge and understanding of a wide range of circumstances. This is especially crucial when the datasets obtained may have limitations in terms of their size. Deep learning models sometimes suffer from overfitting, where they become too specialized in the training instances. To address this, one might include variability into the input photographs to generate new and interesting training examples.

2.4. Convolutional Neural Network (CNN) implementation on tensroflow concept

Convolutional Neural Networks (CNNs) benefit from non-linear mapping by dynamically altering the training weights between neurons. The convolutional, pooling, and fully connected layers are the three fundamental layers of a conventional CNN. The convolutional layer is a vital component in the majority of computations. The system is comprised of kernels, which are made up of weights that acquire visual

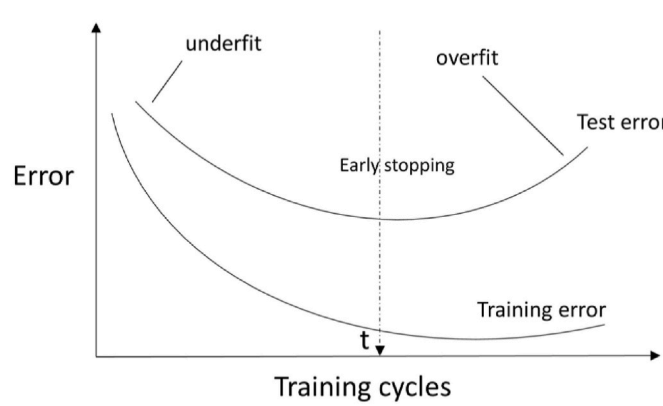

Fig. 14. Error during training and Testing.

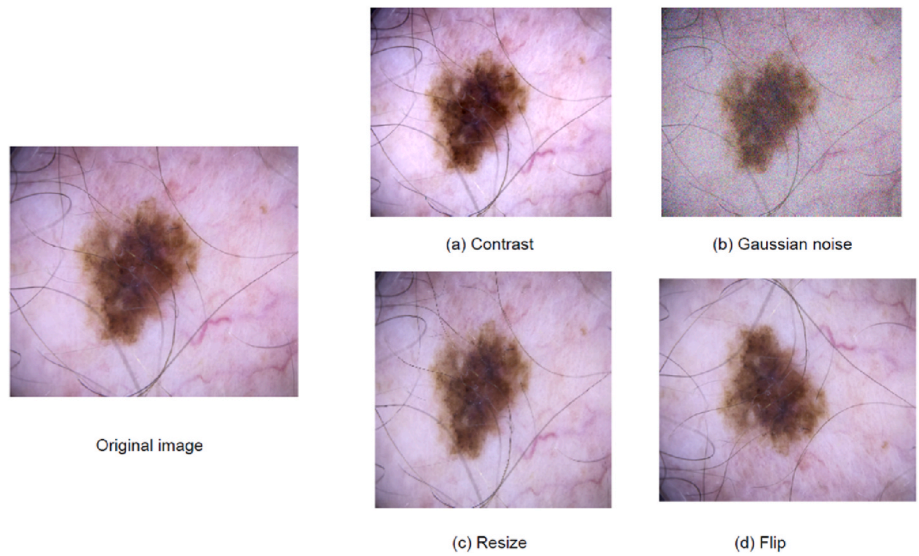

Fig. 15. The disparity between the initial skin lesion image (on the left) and the images subsequent to data augmentation (on the right).

attributes from the input images. The outcome of this layer is a feature map generated by convolving each kernel across the entire image. The main objective of the pooling layer is to decrease the size of the feature map. As a result, it lowers overfitting, decreases the number of training parameters required for successive layers, and introduces non-linearity into the network via a nonlinear activation filter. The fully connected layer is a conventional neural network layer that utilizes the last feature map generated by the previous layer. The feature extractor is comprised of the convolutional and pooling layers, whereas the classifier solely pertains to the fully connected layer.

The ImageNet challenge [2] Regularly employs Convolutional Neural Networks (CNNs), which have been a potent technique for object categorization competitions. CNN models, such as Vgg, are utilized. [3], GoogLeNet [9], and ResNet [11] have outperformed AlexNet [13], They also utilized a Convolutional Neural Network (CNN) architecture and were declared the champions of the annual ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2012. Steve and his colleagues, also known as "et al." in academic writing, have achieved favorable outcomes by employing several Convolutional Neural Network (CNN) architectures to detect and diagnose skin cancer. However, it is difficult to collect a large amount of data on medical activities specifically for the goal of training a Convolutional Neural Network (CNN). Transfer learning is a widely employed method where a model that has been trained for a specific task is utilized to some extent for a new activity. This strategy has been extensively utilized in numerous research studies to address this challenge. Thus, models can be enhanced by utilizing their own dataset after being initialized with the weights derived from the ImageNet dataset [2].

In this study, we utilized TensorFlow's implementation of Convolutional Neural Network (CNN) to store many dimensions of the image dataset. Each dimension is denoted as a feature. Afterwards, we utilize OpenCV to resize the photos to the necessary dimensions, and then save them as NumPy arrays labeled "pos" and "neg". The dimensions and size of the picture datasets are shown by outputting the geometry of each array.

Subsequently, we create an input layer within the model to accept the picture data. Next, we describe the first convolutional layer, which comprises 64 filters, each with dimensions of 3x3. The layer additionally includes a Rectified Linear Unit (ReLU) activation function and is subjected to regularization with a weight decay of 2e-4. Ultimately, this layer is linked to the input layer. We implemented a max-pooling layer to reduce the size of the feature maps from the previous convolutional layer. Then, we applied batch normalization to the resulting output of the max-pooling layer. Subsequently, we define the attributes of the second convolutional layer. The system comprises 128 filters, each measuring 3x3 in size. We utilize the Rectified Linear Unit (ReLU) activation function and employ L2 regularization. This layer is linked to the preceding layer. In addition, we establish an additional max-pooling layer and do batch normalization on the resulting output of this layer.

Ultimately, we utilize the process of flattening to transform the output of the previous layer into a vector with a single dimension. In addition, we create a dense layer with 64 units and apply a sigmoid activation function. This layer receives the flattened output as input and produces the final embedding outcome. A Model object is instantiated from Keras, using the input layer "img_input" and the output layer "embed". This results in the creation of the convolutional base model.

3. Conclusion

Studies have shown that deep learning is a highly efficient state-of-the-art skin lesion classifier for diagnosing skin disorders. The aim of this study was to utilize deep learning techniques to develop a Neural Network model for the examination of dermoscopic images, with the intention of assisting in the prompt detection of skin cancer. The implemented approach significantly improves the accuracy of such detection. The results indicate that using a combination of CNN and GAN models successfully mitigated the issues related to class imbalance, confirming the significance of data augmentation strategies. The high accuracy achieved by our deep learning models in distinguishing between malignant and benign skin lesions demonstrates their potential for real-world application. For instance, the use of these models in mobile health applications could provide individuals, particularly those in rural or underserved regions, with access to preliminary skin cancer assessments. These systems can act as a decision support tool for dermatologists, speeding up the diagnostic process and allowing for earlier intervention.

This study utilized various deep learning architectures. We conducted an analysis of the differences between several categories within a substantial dataset and investigated techniques to improve performance using a smaller subset. It is feasible to improve performance. Our main objective was to use small skin lesion datasets by combining them with advanced deep learning techniques. This study also emphasized the challenges posed by limited datasets and class imbalance in skin cancer detection. The application of transfer learning and augmentation, as presented in this study, confirms their ability to enhance detection accuracy, especially in early-stage melanomas, aligning with the introductory objectives. We have therefore implemented two primary strategies: transfer learning and data augmentation, along with the utilization of Generative Adversarial Networks (GAN). Imbalanced datasets can be mitigated by employing sampling strategies and loss functions.

Finally, detection of skin cancer is critical in reducing mortality rates, especially for aggressive types like melanoma. This study provides a significant step toward improving diagnostic tools using deep learning, with potential implications for real-world clinical applications.

CRediT authorship contribution statement

Olusoji Akinrinade: Writing – original draft. **Chunglin Du:** Supervision.

Declaration of competing interest

We (Olusoji Akinrinade and Chunling Tu) hereby declare that we have no conflicts of interest regarding the publication of the manuscript titled: Skin Cancer Detection Using Deep Machine Learning Techniques in Artificial Intelligence in Medicine. We have no financial or personal relationships with individuals or organizations that could influence our work inappropriately.

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